

Sanath Palakkal

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Profile

MSc Applied Physics student with a specialization in VLSI and a strong foundation in semiconductor devices, electronics, and computational technologies. Experienced in analog IC design, CMOS layout, and Verilog RTL development, with hands-on project experience in Cadence Virtuoso. Passionate about emerging technologies including quantum computing and advanced hardware systems, seeking opportunities to contribute to innovative research and engineering projects.

Domain skills

- **VLSI / IC Design Domain Skills:** CMOS circuit design fundamentals, Digital IC design basics, VLSI fabrication & technology understanding, Physical design basics (layout, DRC, LVS)
- **EDA Tools:** Verilog (RTL design), Cadence Virtuoso, LTspice (circuit simulation), Magic VLSI (layout), Xschem (schematic capture), Silvaco TCAD (device-level simulation)
- **PCB & ECAD Tools:** KiCAD, Eagle
- **Embedded & Programming:** Arduino, C, Python, Verilog

Area of Interest

Digital IC Design, Analog Circuit design, Quantum computing, Verification & Validation, Robotics Fundamentals

Education

MSc	Digital University of Kerala, Trivandrum MSc Applied Physics with specialization in VLSI <i>Coursework:</i> VLSI Technologies, VLSI Physical Design, Digital IC Design, Mixed Signal Design, Neuromorphic Computing, Quantum Mechanics, Statistical Physics	2024 – Ongoing
BSc	Calicut University BSc Physics	2020 – 2023
HSS	GGVHSS Feroke Kerala, India Natural Science	2018 – 2020

Leadership & Responsibilities

Served as General Secretary at Digital University Kerala, coordinating student activities and communication

Academic Projects

VGG-11 Neural Network Hardware Implementation

- Implemented VGG-11 architecture using Verilog HDL
- Performed behavioral simulation and hardware-oriented optimization
- Design targeted for ASIC tape-out
- Analyzed layer-wise operations and resource efficiency

Autonomous Object Detection Car using ESP32

- Built an autonomous obstacle detection robot using ESP32, ultrasonic sensors, and L293D motor driver
- Integrated sensor interfacing, motor control, and power management
- Implemented real-time obstacle avoidance logic in embedded C

Bandgap Reference (BGR) Design in 180nm CMOS (Tape-out Ready)

- Designed and simulated a precision Bandgap Reference using Cadence Virtuoso (SCL 180nm PDK)
- Completed full custom layout and successfully passed DRC and LVS verification
- Design prepared for ASIC tape-out, demonstrating end-to-end analog IC design flow

Certificate & Achievements

- **Letter of Recognition for Leadership & Academic Excellence**, Digital University Kerala (2026)
Recognized by **Prof. Alex P. James** as one of the best performing students in the IMCS Group.
- Attended a Short-Term Training Programme on Custom IC Design and Layout using Standard 180nm PDK organized by National Institute of Technology, Calicut with **IEEE SSCS**.
- **VLSID 2026 Fellowship Recipient** – Selected for the fellowship at the VLSI Design Conference (VLSID 2026), held in **Pune, India**, in recognition of academic merit and interest in VLSI research.